

M13-C3 — H5 Greedy-cautious vs Greedy-brave experimental `greedy_cautious`

H5 run / analysed / not a Bucket 3 claim

1 Status

H5 run / analysed. Experimental `greedy_cautious` vs locked `aamas2025` on all five completed deterministic Bucket 3 AAMAS collections (18 cells). Markdown twin: `h5_greedy_cautious.md`.

Lead (Analysis A): 18/18 outcome and delta match (byte-identical solved deltas); traces isomorphic aside from entailment-logging surface; mean runtime $\approx 1.84\times$ (still sub-second). **Secondary (Analysis B):** large contrasts only vs `baseline_cautious` (nd bundle) — **search strategy**, not the brave $\rightarrow$ cautious flip inside Greedy. Config is experimental (not a published AAMAS method; not `aamas_cautious`). H0–H4b closed. **Not a Bucket 3 claim.** H6 analysed separately (`h6_folding_and_n_ablation.md`); H7a analysed separately (`h7a_relto_vs_sechk.md`); H7b not started.

2 Research question

Under the AAMAS Greedy search bundle on these tables, does switching from brave to cautious change outcomes, deltas, or search paths; and how does Greedy-cautious compare to repository nd-cautious where that arm exists?

3 Main finding (Analysis A — lead)

On all **18** paired cells, `greedy_cautious` (Greedy / mgr / bk / relto + learning_mode(cautious)) matches locked `aamas2025` on **every outcome and every solved delta**. Traces are isomorphic aside from OPTIMUM FOUND vs checking entailment ... OK. Flipping brave $\rightarrow$ cautious **inside the Greedy bundle does not change accepted or rejected theories** on these tables. Runtime is the main observable cost.

No-sol close (H0 *a*; GC):

```
found: alpha_1/1 ... K0, cannot introduce an assumption for
      [b_val_0(A),c_val_0(A)]
* No solution found!
```

c success (H0):

```
folding result: c(A) <- [a_val_1(A),b_val_1(A)]
gen2: extended ABA entails <E+,E-> - using folding
```

4 Setup

Config id `greedy_cautious`; Prolog configs/`greedy_cautious_config.pl` (`aamas2025_config.pl` unchanged). Intervention: `learning_mode` only. Pair integrity: learner-visible inputs byte-identical across AAMAS $\leftrightarrow$ GC (16 non-skipped cells). Timeout 300s preserved; no GC timeouts. Five collections: H0/H1 collider; H2/H2c iso-*D*; H3 BD+lead-*a*.

5 Outcome summary

18/18 outcomes match; all solved deltas byte-identical; 0 final assumptions; all GC runtimes sub-second; mean ratio $\approx 1.84\times$ (range ~ 1.56 – 2.26).

6 *c* and guardrails

H0/H1 *c*: mechanism-aligned string (not automatic causal claim). H2/H3 *c*: distractor-retaining predictive. H1 roots: spurious co-occurrence Horn. Only desirable deterministic target is *c*.

7 Analysis B (secondary)

Hold cautious; compare `greedy+mgr+bk` vs `nd+any+all` (not a single-knob ablation); H0/H3 $n=30$ only.

H0 roots: both no-sol; baseline $\sim 5k$ lines / $\sim 70s$ vs GC ~ 340 lines / $\sim 0.4s$. H0/H3 *c*: different theories (GC Horn vs baseline α -rich) — search bundle. H3 roots/*d*: baseline timeout@300s; GC short no-sol.

8 Semantics vs search-strategy

GC $\equiv$ AAMAS: learning mode does not fork Greedy here. GC vs baseline contrasts: **search bundle**.

9 Boundaries / non-claims

Not a Bucket 3 claim; not “cautious is better”; not mechanism recovery; not an AAMAS-paper cautious method; not H6 or H7 evidence.

10 Next

H5 documentation is complete. H6 is analysed separately (`h6_folding_and_n_ablation.md`); H7a is analysed separately (`h7a_relto_vs_sechk.md`); H7b is not started. Orchestrator / Samuel decides the next move. Still **no Bucket 3 claim**.